PulmoTwin: 48-Hour Respiratory Digital Twin

System Architecture & End-to-End Multi-Modal Data Flow Pipeline

1. INGESTION & COHORT SYNTHESIS LAYER

Static EHR Modality (N=500)
- Synthea COPD/Asthma cohort
- Age (42-84y), Sex, Staging
- Smoking status & pack-years
- Maintenance inhaler adherence

Continuous Wearable Vitals
- 365 Days per patient (182.5k records)
- SpO2, HR, RR, Sleep, Steps
- Individual physiological baselines
- 2-3 Day pre-flare biomarker drift

Atmospheric & Weather (Delhi)
- Open-Meteo CAMS Air Quality
- CPCB NAQI conversion (PM2.5, PM10)
- ECMWF ERA5: Dust, Temp, Humidity
- Delhi Seasonal Pollen Formula (AIIMS)

2. TIME-SERIES FEATURE ENGINEERING & ENVIRONMENTAL FUSION

Biometric Rolling Signals
- Deviation from personal baseline
- 3-day rolling trend (SpO2, HR, RR)
- Exertional drop ratio (Steps/base)
- Nocturnal sleep debt accumulation

Atmospheric Exposure Align.
- Delhi CPCB NAQI mean & 3d max
- Pre-monsoon dust storm peaks
- Diurnal temperature swings (ΔT)
- Rain washout attenuated pollen

Patient-Level Splitting & Scaling
- 80/20 Cohort Split (0 patient leakage)
- 400 Train / 100 Test patients
- Missing sensor dropout augmentation
- Robust 62.98 MB tabular matrix

3. PREDICTIVE AI ENGINE & SHAP ATTRIBUTION

Calibrated XGBoost Classifier
- 48h Advance Flare-Up Forecast
- ROC-AUC: 0.9877 | PR-AUC: 0.9080
- 93.4% Recall @ Optimal Threshold
- 4-Stage Multi-Modal Ablation

TreeExplainer Attribution
- Local SHAP factor attribution
- Sigmoid-shared percentage points
- Baseline (3.6%) + Bars = Risk%
- 9 Clinical actionable factors

Counterfactual Simulation Engine
- Real-time What-If calculations
- HEPA filtration (AQI delta: -150)
- Inhaler compliance booster (+50%)
- Thermal swing mitigation (-8°C)

4. CLINICAL DECISION SUPPORT & DASHBOARD INTERFACE

FastAPI High-Perf Backend
- Asynchronous REST API (Port 8000)
- Dynamic 365d Timeline Replay
- Patient list sorted by 48h risk
- Full OpenAPI / Swagger documentation

3D Twin & Clinical React Dashboard
- Three.js breathing lung twin
- Dynamic tinting & AQI particles
- Dark 'Patient Monitor' telemetry
- 5 Vital tiles with 7-day sparklines

Hospital EHR & Dispatch Workflow
- Grouped multi-day clinical episodes
- Delhi GIS station map with patient pin
- 1-Click nurse action orders
- Workload control: 0.77 alerts/pt/mo